

TOPIC 1 : Machine Learning

1. Can you explain the differences between supervised, unsupervised, and reinforcement learning with examples?
2. What is the purpose of splitting data into training, validation, and test sets?
3. What is the bias-variance tradeoff, and how does it affect model performance?
4. How do you detect and prevent overfitting and underfitting?
5. How does logistic regression differ from linear regression?
6. What are the key advantages of decision trees, and how do random forests improve upon them?
7. Can you explain how gradient boosting algorithms like XGBoost work?
8. What are some limitations of k-Nearest Neighbors?
9. How does the Naïve Bayes classifier work, and in which scenarios is it most useful?
10. What is the kernel trick in SVMs, and why is it important?
11. How do you determine the optimal number of clusters in k-Means?
12. Can you explain the intuition behind PCA?
13. When would you prefer F1 score over accuracy?
14. How do you interpret an ROC curve and AUC score?
15. What regression metric would you choose for a business case where large errors are unacceptable?
16. How do you evaluate a model when your dataset is highly imbalanced?
17. How do you handle missing values in your dataset?
18. When would you choose one-hot encoding over label encoding?
19. Why and when should you scale your features?
20. Can you give an example of an embedded method for feature selection?
21. What is the difference between L1 and L2 regularization?
22. How does dropout help prevent overfitting in neural networks?
23. Can you explain how the Adam optimizer differs from standard SGD?
24. When would you use a learning rate schedule?
25. How does k-fold cross-validation work, and why is it useful?
26. How do you handle imbalanced classes in a classification problem?
27. Can you explain the purpose of using sklearn pipelines?
28. What methods can you use to make your model explainable to non-technical stakeholders?

TOPIC 2 : Deep Learning

1. What is the difference between a perceptron and a multi-layer perceptron?
2. Why do we need activation functions in neural networks?
3. How does the softmax function work and where is it used?
4. Can you explain the process of backpropagation?
5. What are some differences between Adam, SGD, and RMSProp?
6. Why do vanishing gradients occur, and how can they be mitigated?
7. How does the convolution operation extract features from an image?
8. What is the role of stride and padding in convolutional layers?
9. Compare max pooling and average pooling. When would you choose one over the other?
10. What is the main limitation of vanilla RNNs?
11. How does an LSTM cell address the vanishing gradient problem?
12. When would you prefer a GRU over an LSTM?
13. What is self-attention, and how does it work?
14. Why are transformers better suited for long sequences compared to RNNs?
15. What is positional encoding, and why is it necessary in transformers?
16. How does dropout help prevent overfitting?
17. What is batch normalization, and why is it useful?
18. When would you use L1 regularization over L2 regularization?
19. What is the difference between fine-tuning and feature extraction in transfer learning?
20. Can you name some popular pretrained CNN architectures?
21. How would you adapt a pretrained model for a new classification problem?
22. What is the difference between an autoencoder and a variational autoencoder?
23. How do GANs work, and what is the role of the generator and discriminator?
24. What are some common challenges in training GANs?
25. How does batch size affect training and generalization?
26. What are learning rate schedules, and why are they important?
27. How do you detect and handle overfitting in deep learning?

TOPIC 3 : NLP

1. How does tokenization work, and why is it important in NLP?
2. What are stopwords, and when should you remove them?
3. Can you explain stemming vs. lemmatization with examples?
4. What are word embeddings, and how do they capture semantic meaning?
5. Compare Word2Vec, GloVe, and FastText.
6. How do embeddings handle synonyms and polysemy (multiple meanings)?
7. What is the difference between a unidirectional and bidirectional language model?
8. How do n-gram models work, and what are their limitations?
9. How does a transformer-based language model differ from traditional RNN-based models?
10. Why do vanilla RNNs struggle with long-term dependencies?
11. How does an LSTM's gating mechanism work?
12. When would you use a GRU instead of an LSTM?
13. Explain the concept of self-attention.
14. What is the purpose of positional encoding in transformers?
15. How does BERT differ from GPT in architecture and training objectives?
16. How would you build a text classification pipeline from scratch?
17. What evaluation metrics would you use for imbalanced sentiment datasets?
18. How can transfer learning improve text classification?
19. What is NER, and how does it work?
20. How would you handle overlapping or nested entities in NER?
21. Can you give an example of real-world applications of NER?
22. What is the difference between statistical and neural machine translation?
23. How do attention mechanisms improve translation quality?
24. What is beam search, and why is it used in translation models?
25. What are the main differences between extractive and abstractive QA?
26. How does RAG improve retrieval-based NLP tasks?
27. What role does a vector database play in RAG systems?
28. What is BLEU score, and when would you use it?
29. How do ROUGE metrics differ from BLEU?
30. When would F1 score be more informative than accuracy in NLP tasks?

TOPIC 4: Computer Vision

1. What techniques can be used to normalize or standardize images before training?
2. How does data augmentation help in improving model performance?
3. Why might you convert images to a different color space (e.g., HSV)?
4. How do convolutional layers work, and why are they effective for images?
5. What's the role of pooling layers in CNNs?
6. How do CNNs differ from fully connected networks?
7. What is the difference between handcrafted features (e.g., SIFT, HOG) and learned features in deep learning?
8. How do filters in CNNs detect edges, textures, or patterns?
9. What is receptive field, and why is it important?
10. What's the difference between object detection and image classification?
11. How do algorithms like YOLO and Faster R-CNN work at a high level?
12. What is Intersection over Union (IoU), and how is it used in detection tasks?
13. What is the difference between semantic segmentation and instance segmentation?
14. How does U-Net architecture work?
15. What metrics are used to evaluate segmentation models?
16. How does transfer learning benefit computer vision projects?
17. When would you fine-tune all layers vs. only the last few layers?
18. Give an example of using a pretrained model for a custom CV task.
19. How do Vision Transformers differ from CNNs?
20. What is patch embedding in ViTs?
21. When might you prefer a ViT over a CNN?
22. How do GANs work for image generation?
23. What is the role of the discriminator and generator in GANs?
24. What are common issues when training GANs, and how can they be mitigated?
25. How is accuracy different from precision and recall in object detection?